

Reply to the referee report on JHEP_282P_0726

Soft gluon wave function and evolution operator in the CGC at next-to-leading order

Ramkumar Radhakrishnan

Dear Editor,

I thank the referee for a careful reading of the manuscript and for comments that have substantially improved it. Below I reply to each point. All changes are highlighted in colour in the revised manuscript. Equation numbers refer to the revised version unless marked “v1”; in particular Eq. (4.78) of v1 is now Eq. (4.82), and Eq. (4.82) of v1 is now Eq. (4.88).

Comment 1.

It seems that in Sec 4.3.1 one is neglecting a “ t -channel gluon exchange” contribution, where one passes through a three-gluon intermediate state. This is a significant part of the result. It is also curious that the authors do not notice the absence of this contribution in the unitarity relations, which should not hold when it is neglected.

Reply. I agree. Eq. (4.88) contains, in each of the three colour structures, the four-gluon contact interaction and the *instantaneous* gluon exchange in the t -, u - and s -channel. The *propagating* exchange, which is second order in H_{ggg} and passes through an intermediate state, was included only in the s -channel, Eq. (4.90). The propagating t - and u -channel exchanges were missing.

In the new version they are computed explicitly in Sec. 4.3.1, Eqs. (4.91)–(4.94) and Fig. 2. Each channel has two light-cone time orderings, depending on which incoming gluon emits the exchanged gluon. For given external momenta exactly one of them contributes, which is made explicit by Θ -functions. The intermediate energy denominators vanish only for collinear splitting, where the vertex factors vanish linearly, so the transverse integrals are finite. The four corresponding terms are added to the coefficient $\mathbb{D}_2$ in Eq. (5.7).

I also agree about unitarity. For the intermediate state $|g(r)g(r')g(p)\rangle$ the unitarity identity, Eq. (4.15), produces the products $-\mathbb{C}^\dagger(q, r', p) \mathbb{C}(k, r', r)$, in which the internal momentum connects an incoming and an outgoing gluon on different lines. In the old version these products had no counterpart in $\mathbb{D}_2$. They are reproduced exactly by the new terms. The coefficients $\mathbb{B}_3$ and $\mathbb{D}_2$ are fixed completely by the matching to the light-cone wave function, so their unitarity relations are non-trivial checks of the calculation. This is now stated in the text and shown explicitly in Eq. (4.15).

While redoing this part I also corrected the s -channel result, Eq. (4.90): a factor $1/r'^+$ and factors of 2 in the intermediate energy denominator were missing, and the colour factor and one Kronecker delta now agree with the vertex, Eq. (D.3).

Changes. Paragraph after Eq. (4.88); corrected Eq. (4.90); new Eqs. (4.91)–(4.94) and Fig. 2; four new terms in Eq. (5.7); new paragraph with Eq. (4.15) and remark after Eq. (5.12).

Comment 2.

This is related to both the “ t -channel gluon exchange” contribution mentioned above and the contribution already included in (4.82). The $2g \rightarrow 2g$ energy denominator can go to zero and thus needs to be regularized. How this is done should be discussed in the manuscript.

Reply. I agree this must be stated. In the new version this is done in Sec. 3.2 and Appendix E. Even after the momentum-conserving δ -function is used, the denominator $E_i - E_f$ of Eq. (4.88) and of Eqs. (4.91)–(4.94) vanishes for specific kinematics, i.e. for $r = p$, $q = k$. The wave functions are computed using non-degenerate perturbation theory. The free soft gluon states form a continuum,

so the degenerate configurations $E_i = E_f$ make a set of measure zero. The perturbative series is then defined in the usual way by the $i\epsilon$ prescription,

$$E_i - E_f \rightarrow E_i - E_f + i\epsilon \quad \text{at } n\text{-th order,} \quad \epsilon \rightarrow 0^+.$$

This prescription comes from the adiabatic definition of Ω in Eq. (3.7). Here Ω is the Møller operator of scattering theory, and $\Omega|n\rangle$ are the corresponding “in” states. With this, the unitarity relations, Eq. (3.9), hold exactly. A pole at $E_i = E_f$ remains in only two coefficients, viz. $\mathbb{B}_3$ and the ρ -independent term of $\mathbb{D}_2$. There its residue is the on-shell scattering amplitude, and all the other coefficients are free of this pole. Since the pole is simple, $1/(x + i\epsilon) = \text{P}(1/x) - i\pi\delta(x)$ has an integrable principal value, and the δ -function term is an on-shell, anti-Hermitian contribution that does not enter the unitarity relations.

Now consider the transformed Hamiltonian $\Omega^\dagger H \Omega$. The off-diagonal elements between states of different energy cancel, as before. What is left is multiplied by the factor $i\epsilon/(E_i - E_f + i\epsilon)$. This factor equals 1 only on the energy shell, and it goes to zero as $\epsilon \rightarrow 0$ everywhere else. Since the energy shell is a set of measure zero, this leftover part does not contribute to matrix elements between normalizable states. Hence the diagonalization of Sec. 6 holds as stated.

Changes. New paragraphs in Sec. 3.2; Eq. (3.7) rewritten; new Eqs. (3.8)–(3.9); a sentence after Eq. (4.88); Appendix E rewritten in the standard Rayleigh–Schrödinger form.

Comment 3.

At this order in perturbation theory the QCD coupling should run and the NLO result should depend on a scale associated with the coupling constant renormalization. Indeed one has here a β -function coefficient in (4.78), but somehow no running of the coupling constant. This should be addressed in the text.

Reply. I agree this should be discussed. The constant $11/6$ in Eq. (4.82) multiplies the transverse integral $\int d^2\tilde{p}/\tilde{p}^2$, which is scaleless and so vanishes in dimensional regularization. It vanishes because its UV and IR divergences cancel, not because it is finite:

$$\mu^{2\epsilon} \int \frac{d^{2-2\epsilon}\tilde{p}}{\tilde{p}^2} = \pi \left(\frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}} \right) = 0.$$

The UV part of the $11/6$ term is $-\frac{g^2}{16\pi^2} \frac{\beta_0}{\epsilon_{\text{UV}}}$ with $\beta_0 = \frac{11}{3}N_c$. This is the one-loop gluon field-strength renormalization. In light-cone gauge it also fixes the renormalization of the coupling, $Z_g = Z_3^{-1/2}$ [Leibbrandt, Rev. Mod. Phys. 59 (1987) 1067], which is why the β -function coefficient appears here.

The normalization N therefore contains no renormalization scale at $\mathcal{O}(g^2)$. The UV divergence is absorbed into the renormalized coupling $g(\mu)$. The scale enters through the coupling: the $\mathcal{O}(g)$ coefficients $\mathbb{A}$ and $\mathbb{C}$ are tree level and are functions of $g(\mu)$, and their one-loop, $\mathcal{O}(g^3)$, corrections [Lublinsky and Mulian, JHEP 05 (2017) 097] carry the compensating $\log \mu^2$ with a coefficient $\propto \beta_0$. Together they turn the fixed coupling of the Weizsäcker–Williams amplitude into a running coupling at a scale set by the gluon transverse momentum, as in the running-coupling BK/JIMWLK kernels [Balitsky, Phys. Rev. D 75 (2007) 014001; Kovchegov and Weigert, Nucl. Phys. A 784 (2007) 188]. Computing the $\mathcal{O}(g^3)$ terms is beyond the $\mathcal{O}(g^2)$ evolution operator constructed here.

Changes. Text after Eq. (4.82) and new Eqs. (4.83)–(4.85); a sentence after Eq. (5.2); the corresponding remark after Eq. (6.8) and in the Conclusions; three new references.

Comment 4.

In Sec 3.1 it is stated that the “coefficients are not ordinary c-number functions but operator quantities acting in the valence Hilbert space”. [...] The statement about noncommuting operator coefficients only applies to the soft gluon part of the Hilbert space, not the full one. The fact that the states in e.g. eq (3.2) are states in only the soft subspace should be emphasized more in the text.

Reply. I agree. In the new version this is emphasized in Sec. 3.1. The expansion in Eq. (3.2) is an expansion in the Fock basis of the soft subspace $\mathcal{H}_{\text{soft}}$ only. The full Hilbert space $\mathcal{H}_{\text{soft}} \otimes \mathcal{H}_{\text{val}}$ is an ordinary complex vector space, and in a complete basis $|n\rangle_{\text{soft}} \otimes |v\rangle_{\text{val}}$ all coefficients are c -numbers. Since the valence part is not expanded, the coefficients of the soft Fock states are functionals of the valence charge density, i.e. operators on $\mathcal{H}_{\text{val}}$, and their ordering must be kept because of Eq. (2.19). All states of the form (3.2), and the vacuum $|0\rangle \equiv |0\rangle_{\text{soft}} \otimes |v\rangle$, should be understood in this sense. The same remark is repeated where Ω is expanded in Sec. 4.

Changes. Rewritten paragraph after Eq. (3.2); sentence after Eq. (4.1).

Other changes. In a complete re-check of the manuscript I also corrected typographical errors in a number of equations: signs, indices, integration measures, and missing Hermitian-conjugate terms in Secs. 2, 4, 5 and 6 and in Appendices A–C. I also stated the plus-momentum window that regularizes the gluon-number-preserving transitions in Sec. 4.2. In addition, I clarified that the $\mathcal{O}(g)$ components of $\Omega|gg\rangle$ and $\Omega|ggg\rangle$ contribute to the normalization and that $\mathbb{N}$ is fixed by unitarity, and stated the terms that cancel in the diagonalization, after Eq. (6.7). The Introduction now refers to pure Yang–Mills theory rather than QCD, consistent with the content of the paper.

I hope the revised manuscript is now suitable for publication in JHEP.

Sincerely,
Ramkumar Radhakrishnan